

Mainul Hossain

 Github |  Website |  LinkedIn |  hossainmainul83@gmail.com |  Tel: +8801795897030

SUMMARY

Versatile and solutions-driven Site Reliability Engineer with a strong background in cloud infrastructure, automation, and systems administration. Extensive expertise in managing and optimizing Microsoft Azure and AWS environments, with hands-on experience in Docker, Kubernetes, and infrastructure-as-code practices. Skilled in scripting with Python, Bash, PowerShell, and Ruby for automation, monitoring, and API integrations.

Built and maintained AI-assisted reliability tooling (LLM + ML forecasting/anomaly detection) with real-time dashboards and end-to-end observability. Research interests include machine learning, deep learning, and explainable AI (XAI), with practical exposure to cybersecurity and system reliability. Adept at collaborating across teams, adapting quickly to evolving technologies, and driving operational excellence in both cloud and hybrid infrastructures.

WORK EXPERIENCE

Site Reliability Engineer - [Optimizely Inc.](#)

Jun 2025 - Present

Part of **Reliability/Platform Engineering** Team

- Created an AI-powered infrastructure monitoring tool and Agent Prompt Optimization workflow that analyzes web application performance for predicting Black Friday readiness, with built-in PowerShell scripts for automated performance data collection and actionable recommendations for smart server scaling.
- Implemented a Terraform-managed Datadog Workflow that auto-restarts the Datadog agent on Elasticsearch nodes via Azure RunCommand, with VM/VMSS detection, auditable success/failure events, and interval re-notification escalation to on-call engineers.
- Contributed to maintaining the reliability, scalability, and performance of Optimizely's cloud-native SaaS infrastructure through proactive monitoring and observability implementations.
- Participated in on-call and support rotations, resolving production issues and improving service reliability.
- Implemented well-scoped automation tasks within internal systems, including CI/CD enhancements and containerized workflow optimization, reducing manual operational overhead.
- Built and maintained an observability stack using Datadog (monitoring, logging, tracing) to ensure reliability and performance across production environments.
- Collaborated across SRE, Operations, and Engineering teams to support service delivery, application lifecycle operations, and define SLOs/SLIs for critical services.
- Implemented Azure automation for .NET Core proactive monitoring and diagnostics, enhancing system observability and reducing incident response time.
- Continued to expand knowledge in system lifecycle management, infrastructure-as-code practices, and secure deployment workflows in AWS and Azure environments.

Managed Services Engineer - **Optimizely Inc.**

Mar 2024 - Jun 2025

Part of **Strategic Support Solutions** Team

- Deployed containerized Python automation solutions using Docker/Kubernetes for auditing and cost optimization processes.
- Implemented infrastructure-as-code practices for Azure service optimization, reducing operational costs by 25%.
- Developed Python and Bash automation scripts for API integrations, data extraction, and troubleshooting workflows.
- Enhanced CI/CD pipelines using GitHub Actions for microservices deployment, reducing deployment time by 30%.
- Built monitoring and alerting solutions using Azure KQL and Datadog for distributed system observability.
- Administered Linux-based systems and managed AWS/Cloudflare CDN services for high-availability architectures.
- Maintained incident management processes and provided L2/L3 support for production system outages.

Associate Managed Services Engineer - **Optimizely Inc.**

Sep 2022 - Mar 2024

Part of **Incident Management** Team

- Led incident response during high-volume traffic surges and major system outages, ensuring rapid detection and resolution.
- Managed production deployments and database changes with zero-downtime deployment strategies.
- Administered F5 load balancers and network security configurations for distributed systems.
- Served as Azure Administrator for DXP platform, maintaining 99.5% uptime across multiple regions.
- Implemented standardized incident management procedures, improving escalation processes and documentation.

Information Technology Officer - **University Of Scholars**

Mar 2022 - Sep 2022

Part of **Information Technology** Team

- Designed and maintained Linux/Windows server infrastructure for university database and web systems.
- Deployed Windows Server domain controllers and implemented organization-wide monitoring solutions.
- Managed LAN/WAN network architecture and IP device configurations for campus connectivity.

EDUCATION

May 2015 - Jan 2022 B.Sc in Computer Science - **BRAC University**

(GPA: 3.14/4.0)

TECHNICAL SKILLS

Programming Languages -	Python, Ruby, JavaScript, Java, C#, PHP
Cloud Platforms -	Microsoft Azure, AWS, Google Cloud Platform
Infrastructure-as-Code -	Terraform, Ansible, Azure DevOps
Scripting & Automation -	Python, Bash, PowerShell
Containerization -	Docker, Kubernetes, Container Orchestration
AI/ML & LLM -	Prompt Engineering, LLM integration, Agent tool design, Ollama, scikit-learn, LightGBM, Prophet
Observability Tools -	Datadog, Prometheus, Grafana, Loki, Promtail, Azure KQL, ELK Stack, New Relic, Pingdom
CI/CD Platforms -	GitHub Actions, Azure DevOps, Octopus Deploy
Operating Systems -	Linux (Ubuntu, CentOS, RHEL), Windows Server
Databases -	PostgreSQL, MySQL, Azure SQL Database
Storage/Cloud Services -	Amazon S3, Azure Blob Storage
Network & Security -	F5 Load Balancers, Cloudflare CDN, Network Configuration

KEY PROJECTS

Infrastructure Self-Healing Agent (AI/ML + LLM)

[GitHub](#)

- Built a production-oriented auto-remediation platform for containerized services with a real-time React dashboard and WebSocket updates (Flask-SocketIO).
- Implemented optional ML for ops signals: anomaly detection (Isolation Forest), forecasting (Prophet), and failure-risk prediction (LightGBM) with artifacts stored in PostgreSQL.
- Integrated a local LLM workflow (Ollama; default Llama 3.2) for incident RCA-style summaries, suggested remediation, and chat-based operator assistance.
- Packaged for Docker Compose and Kubernetes (Helm/Kustomize) with Prometheus metrics and optional Grafana/Loki logging stack.

.NET Core Monitoring & Diagnostics (Azure App Service)

[GitHub](#)

- Created a master Bash runbook tool to monitor thread count, response time, and outbound connections for .NET Core workloads on Linux.
- Automated dump/trace collection using dotnet-dump, dotnet-trace, and dotnet-counters with safe locking to prevent duplicate captures.
- Added Azure Blob upload automation (AzCopy) with retries and instance-aware log rotation for production troubleshooting workflows.

Cloudflare DNS Records Export Tool (Parallel Extraction)

[GitHub](#)

- Built a Python exporter to inventory DNS records across Cloudflare zones, filtered by account, with pagination and retry logic.
- Improved performance using ThreadPoolExecutor parallelism and Docker-friendly output defaults for CI/CD and container runs.

- Automated extraction of Cloudflare custom firewall rules across all zones with action-based filtering (e.g., skip/block/challenge).
- Produced audit-ready CSV reports using parallel zone processing and resilient API retry/backoff behavior.

- Built an asyncio/aiohttp tool for concurrent hostname checks using the Pingdom API with structured CSV reporting for ops workflows.

CERTIFICATIONS & AWARDS

Microsoft Certified: Azure Administrator Associate Microsoft Corporation - Credential ID MS0995597094	Issued Jun 2024
Spot Award Optimizely Spot Award for "Never Stop Improving"	Issued Aug 2023
Spot Award Optimizely Spot Award for "Getting it done together"	Issued Apr 2023
Introduction to Programming Using Java LICT Project, ICT Division, Government of Bangladesh - Credential ID G047514	Issued Mar 2018